Tristan Tan

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Education

University of California, Berkeley — Berkeley, CA
B.A. Data Science and Economics (Intended); GPA: 3.88 — *Aug 2025 – May 2029*

Coursework: Data Science (Foundations, Computational Structures), Probability & Statistics, Linear Algebra
Upcoming: Berkeley Global Internships, Data Science Placement, London (Spring 2027)

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript, R, HTML/CSS
Machine Learning & Data: pandas, scikit-learn, XGBoost, LightGBM, PyTorch, Sentence Transformers, GeoPandas
Visualization & Apps: Streamlit, Tableau, Matplotlib, Plotly, Altair
Web & Infrastructure: React, Next.js, React Three Fiber, FastAPI, PostgreSQL, Supabase, OAuth 2.0, Git, Vercel

Experience

Undergraduate Research Assistant — Sep 2026 – Present
Archaeological Research Facility, UC Berkeley — *Berkeley, CA*

- Lead development of a 3D annotation platform (Next.js, TypeScript, React Three Fiber) anchoring 314 scholarly annotations to 607 regions on 6 Egyptian coffin models, coordinating student researchers and designers
- Implemented mesh raycasting and bidirectional spatial selection linking 3D hieroglyphic inscriptions to scholarly translations and text records
- Built a Supabase/PostgreSQL backend with row-level security and server-side data fetching, deployed on Vercel, replacing manual XML lookup

Data Research Intern — Sep 2026 – Present
Ten Strands — *Berkeley, CA*

- Built a hybrid information-retrieval pipeline (BM25, dense Sentence Transformer embeddings, CrossEncoder reranking) to surface sustainability evidence across 10,000+ California K–12 district documents
- Evaluated lexical and neural retrieval against human-validated evidence using Recall@K and MRR, improving Recall@5 by 35% and reducing manual source-discovery time by 76%

Data Science Intern — May 2026 – Aug 2026
IDX Exchange — *Remote*

- Built an end-to-end home valuation pipeline (Python, pandas, scikit-learn, XGBoost) on 306K California MLS sales, with leakage auditing, feature engineering, tuning, and chronological backtesting
- Achieved 7.7% median error and 85.5% of predictions within 20% on 11,900 held-out sales, 53% lower error than a linear baseline, after geospatial features (GeoPandas join across 680 school districts) cut error 15%
- Shipped a Streamlit app with per-prediction explanations on a 4 MB XGBoost model, 98% smaller than the forest

Undergraduate Research Apprentice — Aug 2025 – May 2026
Lucarelli Photogrammetry Lab, UC Berkeley — *Berkeley, CA*

- Automated a Python photogrammetry pipeline on the Agisoft Metashape API, processing 1,000+ images into 3D reconstructions and cutting manual processing time 40% (30 to 18 min per model)
- Built a reproducible experiment framework and Ridge models on SfM and image-overlap graph features, achieving up to 13.7% lower cross-validated RMSE than a reprojection-error baseline

Projects

Quantitative ML Stock Analytics Dashboard | *Python, pandas, scikit-learn, Streamlit, yfinance*

- Built a live-market Streamlit dashboard benchmarking SVR and Random Forest with time-series-aware splits

Teaching

Computer Science Mentor — Aug 2025 – Present
Berkeley ANova — *Berkeley, CA*

- Teach weekly Python programming sessions to 100+ high school students